Contents

MEDIALERT | PRODUCT REQUIREMENTS DOCUMENT & TECHNICAL SPECIFICATION (PRD)

Project Name: MediAlert (Pill Reminder & Medication Management App)
Document Version: 2.0 (Offline-First Architecture & Design Language Overhaul)
Date: August 2026
Document Status: Approved / Ready for Engineering
Platform Target: Android & iOS (Cross-Platform Flutter)
Primary Architecture: Clean MVVM (Provider + BaseViewModel) with GetIt Service Locator
Data Storage Model: 100% Local Device Database (Isar / Local Storage) — Offline-First
Design System: Clinical Humanist (Corporate Modern + Humanist Warmth)

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1. Executive Summary & Vision

1.1 Product Overview

MediAlert is a high-reliability, offline-first mobile application designed to empower patients, elderly users, and caregivers to manage complex medication regimens with confidence. MediAlert solves the pervasive problem of medication non-adherence by providing precise audio/visual alarm reminders, visual pill photo verification, explicit meal-relation guidance (e.g., "Take after food"), inventory tracking, and comprehensive adherence analytics.

1.2 The Offline-First Mandate

Unlike conventional health applications that rely on mandatory cloud synchronization, user logins, and remote backends, MediAlert operates **strictly offline-first**: - **Zero Remote Dependencies:** All medicine schedules, intake logs, photos, and emergency contacts are stored locally on the user's physical device. - **Instantaneous Reliability:** Reminders and alarms execute natively through hardware-level exact alarm schedulers, immune to network dropouts or server downtime. - **Absolute Patient Privacy:** Medical data never leaves the device without the user's explicit consent via local JSON/PDF export.

1.3 Core Product Objectives

1. **Zero-Failure Reminder Engine:** Deliver actionable, persistent audio/visual alarms with visual pill identification at scheduled times.
2. **Visual Intake Guidance:** Display actual medication photos and clear intake instructions (with meals, empty stomach, etc.) directly on the active alarm screen to prevent dosage errors.
3. **Frictionless Intake Logging:** Enable 1-tap intake confirmation ("Take", "Skip with reason", "Snooze") directly from the dashboard and alarm screens.
4. **Actionable Adherence Analytics:** Generate accurate mathematical adherence reports, streak metrics, visual charts, and printable medical PDF reports for clinical visits.
5. **Data Sovereignty:** Enable full local JSON backup and restore, plus biometric/PIN app lock security.

1.4 Non-Goals for Version 1.0 (MVP)

- **No Cloud User Accounts / Authentication:** No remote server authentication, OAuth, or cloud database sync.
- **No Direct Electronic Health Record (EHR) Integrations:** No direct HL7/FHIR API connections to hospital systems.
- **No Multi-Device Real-Time Sync:** No cross-device cloud websockets or distributed databases.
- **No Integrated E-Commerce Pharmacy:** No automated prescription refill ordering or payment gateway integrations.

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2. Target User Personas & Medical Use Cases

Persona	Demographics & Context	Core Pain Points	MediAlert Solution
Forgetful & Busy Individuals	Working professionals (25–50) taking daily vitamins or temporary prescriptions	Forgetting morning doses; getting distracted by meetings; confusing similar pills	Persistent high-priority alarms, 10-minute quick snooze, and clear pill photo identification.
Elderly Users & Low-Vision	Seniors (65+) managing daily maintenance medications (hypertension, diabetes)	Small fonts; confusing multi-step interfaces; missing alarms when phone is idle	High-contrast UI (Inter font, large text), Clinical Humanist color tokens, 48x48px min touch targets, full-screen alarm override.
Chronic Care Patients	Individuals with multi-drug regimens (Polypharmacy: 5+ pills daily)	Complex intake timing (before meal, after meal, bedtime); drug confusion; tracking adherence	Categorized time slots (Morning, Afternoon, Evening, Night), meal relation tags, inventory refill alerts, adherence history.

Persona	Demographics & Context	Core Pain Points	MediAlert Solution
Short-Term Prescription Patients	Patients prescribed a 7–14 day antibiotic or post-surgery course	Stopping antibiotics prematurely; forgetting mid-day doses	Fixed duration scheduling (Start Date to End Date) with automatic completion celebration and course history log.
Caregivers & Guardians	Family members caring for aging parents or children	Uncertainty if patient took medicine; need verifiable logs for doctor consultations	Clean chronological intake history logs, adherence percentage calculations, and one-tap printable medical PDF exports.

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3. Medical Domain Rules & Adherence Logic

3.1 Medication Intake Statuses

```
stateDiagram-v2
    [*] --> Scheduled: Alarm Configured
    Scheduled --> ActiveAlarm: Scheduled Time Reached
    ActiveAlarm --> Taken: User taps "Mark as Taken"
    ActiveAlarm --> Snoozed: User taps "Snooze" (10m)
    Snoozed --> ActiveAlarm: Snooze Interval Expires
    ActiveAlarm --> Skipped: User taps "Skip" + Reason
    ActiveAlarm --> Missed: Grace Period (30m) Unattended
    Scheduled --> Taken: Early Intake via Dashboard
    Scheduled --> Skipped: Early Skip via Dashboard
```

Status	Definition	Trigger Mechanism	Adherence Calculation Impact
Scheduled / Pending	Dose is upcoming for today and has not yet triggered.	Initial state created by scheduler.	Neutral (not counted in final adherence until time passes).
Taken	User confirmed taking the exact dose.	User taps “Mark as Taken” on Alarm or Dashboard.	**Increments** Adherence Rate.
Skipped	User intentionally chose not to take the dose (e.g., doctor advised, fasting, nausea).	User taps “Skip” and selects an optional reason.	**Excluded** from adherence denominator (does not penalize patient).
Missed	Alarm was dismissed without intake or left unattended past the 30-minute grace window.	System auto-logs after 30-minute timeout.	**Decreases** Adherence Rate.

Status	Definition	Trigger Mechanism	Adherence Calculation Impact
Snoozed	User deferred reminder for a temporary interval.	User taps “Snooze” (default 10 min, configurable).	Temporarily reschedules trigger; remains pending.

3.2 Mathematical Adherence Calculations

To ensure clinical accuracy for medical reports:

1. **Total Scheduled Doses:**

$$\text{Total Scheduled} = \text{Taken} + \text{Skipped} + \text{Missed}$$

2. **True Adherence Rate (%):**

$$\text{Adherence Rate (\%)} = \left(\frac{\text{Taken Doses}}{\text{Total Scheduled Doses} - \text{Skipped Doses}} \right) \times 100$$

(Note: Skipped doses with valid patient reasoning are excluded from the denominator to reflect true therapeutic compliance).

3. **Missed Dose Percentage (%):**

$$\text{Missed Percentage (\%)} = \left(\frac{\text{Missed Doses}}{\text{Total Scheduled Doses}} \right) \times 100$$

4. **Consecutive Streak Counter:**
   - Increments by $+1$ for every calendar day where 100% of scheduled non-skipped doses are marked **Taken**.
   - Resets to 0 if any scheduled dose becomes **Missed**.

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4. Design Language System: Clinical Humanist

MediAlert employs the **Clinical Humanist** design system. It combines the rigorous clarity of medical-grade software with human warmth, soft visual aesthetics, and accessible ergonomics.

```
+--------------------------------------------------------------------------+
|                       CLINICAL HUMANIST DESIGN SYSTEM                     |
|                                                                          |
|  [ Calm Teal Primary ]    [ Success Green ]     [ Alert Red ]            |
|      #00685F                  #006E2F              #B61722               |
|                                                                          |
|  [ Typography ]    Inter (32sp / 24sp / 20sp / 16sp / 14sp / 12sp)      |
|  [ Elevation ]     L0: #F8FAFC  |  L1: #FFFFFF (Cards)  |  L2: Modals   |
|  [ Corner Radius ] Standard: 8px (sm/md)  |  Medication Cards: 24px (xl)|
|  [ Spacing Grid ]  4px Baseline Grid | 16px Gutter | 48x48px Touch Area |
+--------------------------------------------------------------------------+
```

4.1 Color Palette & Design Tokens

Token Name	Hex Code	Semantic Role
primary	#00685F	Calm Medical Teal — Main app bars, active tabs, primary action buttons, brand anchors.
primary-container	#008378	Elevated teal containers and header cards.
on-primary	#FFFFFF	Text and icons displayed on top of primary colors.
secondary	#006E2F	Vitality Green — "Mark as Taken", 100% adherence indicators, positive streak badges.
secondary-container	#6BFF8F	Light success container for highlights and badge backgrounds.
tertiary / error	#B61722	Alert Coral/Red — "Missed Dose", emergency contacts, delete medication, low stock.
tertiary-container	#FFDAD6	Soft red container for missed dose cards and urgent alerts.
background	#F8FAFC	App canvas background (soft off-white to eliminate eye strain).
surface	#FFFFFF	Base card surface, dialog surface, bottom sheet background.
surface-container	#ECEEF0	Tiered neutral surface for chips, disabled states, and divider tracks.
textPrimary	#191C1E	High-contrast charcoal for titles, medication names, and vital data points.
textSecondary	#3D4947	Muted slate for dosages, time stamps, and secondary metadata.
outline	#E2E8F0	Subtle 1px structural borders on cards and text fields.

4.2 Typography Hierarchy (Font: Google Fonts Inter)

All typography strictly integrates with `flutter_screenutil` (`.sp`) and adheres to WCAG AA contrast ratios ($\geq 4.5 : 1$):

Style Name	Size / Weight	Line Height	Letter Spacing	Usage in UI
`Display Large`	`32.sp` / Bold (`700`)	`40.sp`	`-0.02em`	Daily Adherence Percentage, Alarm Clock Time.
`Headline Medium`	`24.sp` / SemiBold (`600`)	`32.sp`	`-0.01em`	Screen Titles (Dashboard, Reports, Add Med).
`Headline Small`	`20.sp` / SemiBold (`600`)	`28.sp`	`0.0em`	Medication Name on Timeline Card & Alarm.
`Body Large`	`18.sp` / Regular (`400`)	`28.sp`	`0.0em`	Intake instructions ("Take with a full glass of water").
`Body Medium`	`16.sp` / Regular (`400`)	`24.sp`	`0.0em`	Form inputs, dialog body text, list descriptions.
`Label Medium`	`14.sp` / SemiBold (`600`)	`20.sp`	`+0.01em`	Time badges, Dosage labels ("500 mg - 1 Tablet").
`Label Small`	`12.sp` / Medium (`500`)	`16.sp`	`+0.02em`	Category chips ("With Food", "Antibiotic", "Daily").

4.3 Shapes, Elevation & Spacing Principles

- **Border Radii:**
  - Standard Buttons & Text Fields: `8px` (`0.5rem` / `AppRadius.sm`)
  - Hero Medication Cards: `24px` (`1.5rem` / `AppRadius.xl`) for an inviting "holding" feel.
  - Status Indicators & Badges: Fully pill-shaped circular radii (`9999px`).
- **Elevation Layers:**
  - **Level 0 (Canvas):** `#F8FAFC` — Flat base.
  - **Level 1 (Cards):** `#FFFFFF` with ambient shadow `0px 4px 20px rgba(0, 0, 0, 0.04)`.
  - **Level 2 (Active Alarms & Dialogs):** `#FFFFFF` with pronounced shadow `0px 10px 30px rgba(0, 0, 0, 0.12)`.
- **Layout Grid & Touch Ergonomics:**
  - 4px baseline rhythm (`xs: 4px`, `sm: 8px`, `md: 16px`, `lg: 24px`, `xl: 32px`).
  - Strict minimum interactive touch target: 48×48 **px** for all buttons, checkboxes, and icons to assist tremors/elderly hands.

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5. Software Architecture & Offline-First Engineering

5.1 Architectural Overview (Adapted from Mentor Guide)

The codebase follows **Clean MVVM (Model-View-ViewModel)** with dependency injection powered by **GetIt**. Source code is cleanly bifurcated into a `core` layer (pure logic, services, models) and a ui layer (screens, custom widgets, state providers).

```
lib/
├── main.dart                # Bootstrap, Service locator init, Notification channels
├── app/
│   ├── app.dart             # MaterialApp, ScreenUtilInit (375x812), Route generator
│   ├── locator.dart            # GetIt singleton registrations for all core services
│    ├── routes.dart                # Centralized Named Route definitions
│    └── theme.dart                 # ThemeData matching Clinical Humanist tokens
├── core/
│   ├── constants/            # AppColors, AppTextStyles, AppSpacing, AppRadius, AppStrings
│    ├── enums/                 # ViewState, MedicineStatus, MealType, FrequencyType
│   ├── models/              # Data entities (Medicine, ReminderTime, DoseLog, Contact)
│   ├── services/             # DatabaseService (Isar), NotificationService, AlarmService,
│   │                        # LocalStorageService (SharedPreferences), PermissionService
│   └── view_model/             # BaseViewModel (safe setState, ViewState management)
└── ui/
   ├── custom_widgets/        # Reusable UI atoms (AppButton, PillCard, ProgressRing, etc.)
    └── screens/                    # Feature screens with co-located ViewModels:
        ├── splash/                 # SplashScreen & SplashProvider
        ├── onboarding/             # OnboardingScreen & OnboardingProvider
        ├── dashboard/              # DashboardScreen & DashboardProvider
        ├── medicine/               # AddMedicineScreen & AddMedicineProvider
        ├── alarm/                  # ActiveAlarmScreen & AlarmProvider
        ├── history/                # HistoryScreen & HistoryProvider
        ├── reports/                # ReportsScreen & ReportsProvider
        ├── emergency/              # EmergencyScreen & EmergencyProvider
        └── settings/               # SettingsScreen & SettingsProvider
```

5.2 Mentor Guide Adaptation: Transitioning from Remote REST to 100% Local Database

In generic project guides, `DatabaseService` is frequently misnamed and wired to remote REST/Dio HTTP clients. **In MediAlert, this is formally replaced by a true local database engine:**

Architectural Layer	Remote API Pattern (Generic Guide)	MediAlert Offline-First Implementation
`DatabaseService`	Wraps REST API / Dio endpoints	Wraps **Isar Database (NoSQL/Relational local SQLite engine)** for direct local queries.
Data Latency	200ms – 2000ms network round-trip	$<$ 5ms instantaneous disk I/O.
Availability	Requires 4G/5G/WiFi connectivity	**100% offline uptime** (works in airplane mode, remote areas).
Media Attachments	S3 / Cloud Bucket URLs	Local file system sandboxed paths via `path_provider`.
User Identifiers	JWT / Bearer tokens in headers	Local device auto-incrementing integer IDs / UUIDs.

5.3 State Management & ViewModel Lifecycle

Every screen uses a co-located `*_provider.dart` extending `BaseViewModel`:

```dart
// Core BaseViewModel Pattern
class BaseViewModel extends ChangeNotifier {
  ViewState _state = ViewState.idle;
  ViewState get state => _state;
  bool _isDisposed = false;

  void setState(ViewState viewState) {
    if (_isDisposed) return;
    _state = viewState;
    notifyListeners();
  }

  @override
  void dispose() {
    _isDisposed = true;
    super.dispose();
  }
}
```

- **Rule of Thumb:** Always invoke `setState(ViewState.busy)` before async operations and `setState(ViewState.idle)` upon completion.
- Screens bind via `ChangeNotifierProvider` and consume state via `Consumer<T>`.

5.4 Logging Conventions

Raw `print()` and `debugPrint()` are strictly forbidden in production business logic. All classes use `logger` with formatted class and method tags:

```dart
final log = CustomLogger(className: '@DashboardProvider');
log.i('@loadTodayTimeline: Loaded 6 doses for today');
log.e('@markAsTaken: Failed to update dose log', error, stackTrace);
```

---

6. Local Database Schema & Data Models (Isar)

All schemas are compiled with `isar_generator` for type-safe, ACID-compliant local transactions.

```
erDiagram
    MEDICINE ||--o{ REMINDER_TIME : "has scheduled"
    MEDICINE ||--o{ DOSE_LOG : "records"
    REMINDER_TIME ||--o{ DOSE_LOG : "triggers"
    USER_SETTINGS ||--o{ EMERGENCY_CONTACT : "manages"

    MEDICINE {
        int id PK
        string name
        double dosageValue
        string dosageUnit
        string pillImageLocalPath
        string formFactor
```

```
        string colorHex
        int mealTypeEnum
        int frequencyTypeEnum
        list specificDaysOfWeek
        int intervalHours
        date startDate
        date endDate
        bool isOngoing
        string doctorName
        string prescriptionNotes
        int currentStock
        int lowStockThreshold
        bool isRefillAlertEnabled
        datetime createdAt
    }

    REMINDER_TIME {
        int id PK
        int medicineId FK
        int hour
        int minute
        bool isActive
        string soundRingtone
        bool isVibrationEnabled
    }

    DOSE_LOG {
        int id PK
        int medicineId FK
        int reminderTimeId FK
        datetime scheduledDateTime
        datetime actualTakenDateTime
        int statusEnum
        string skipReason
        string notes
        datetime createdAt
    }

    EMERGENCY_CONTACT {
        int id PK
        string fullName
        string relationship
        string phoneNumber
        bool isPrimary
    }

    USER_SETTINGS {
        int id PK
        bool isBiometricEnabled
        string pinHash
        string themeMode
        int snoozeDurationMinutes
        int gracePeriodMinutes
```

```
}
```

6.1 Entity Specification: `Medicine`

- `id` (Id): Auto-incrementing primary key.
- `name` (String): Commercial or generic drug name (e.g., "Amoxicillin").
- `dosageValue` (double): Numerical strength (e.g., `500.0`).
- `dosageUnit` (String): Unit of measure (`"mg"`, `"ml"`, `"Tablet"`, `"Capsule"`, `"Drops"`, `"Puff"`).
- `pillImageLocalPath` (String?): Sandboxed local file URI to the captured photo.
- `formFactor` (String): Physical shape identifier (`"tablet"`, `"capsule"`, `"syrup"`, `"injection"`, `"inhaler"`).
- `colorHex` (String): UI color accent for the medication card.
- `mealType` (Enum: `MealType`): `beforeMeal`, `afterMeal`, `withMeal`, `emptyStomach`, `noRelation`.
- `frequency` (Enum: `FrequencyType`): `daily`, `specificDays`, `interval`, `asNeeded`.
- `specificDaysOfWeek` (List): Days active (1 = Monday … 7 = Sunday).
- `intervalHours` (int?): E.g., every 8 hours.
- `startDate` (DateTime) & `endDate` (DateTime?): Course duration.
- `isOngoing` (bool): True for permanent maintenance therapy.
- `doctorName` (String?) & `prescriptionNotes` (String?): Special medical remarks.
- `currentStock` (int) & `lowStockThreshold` (int): Inventory counter.
- `isRefillAlertEnabled` (bool): Triggers alert when `currentStock <= lowStockThreshold`.

6.2 Entity Specification: `ReminderTime`

- `id` (Id): Primary key.
- `medicineId` (int): Foreign key link to parent `Medicine`.
- `hour` (int: `0..23`) & `minute` (int: `0..59`): Exact 24-hour time trigger.
- `isActive` (bool): Master toggle for the reminder slot.
- `soundRingtone` (String): Selected alarm tone asset path or system sound URI.
- `isVibrationEnabled` (bool): Haptic motor toggle.

6.3 Entity Specification: `DoseLog`

- `id` (Id): Primary key.
- `medicineId` (int): Foreign key link to `Medicine`.
- `reminderTimeId` (int?): Foreign key link to triggering reminder.
- `scheduledDateTime` (DateTime): Exact scheduled timestamp.
- `actualTakenDateTime` (DateTime?): Timestamp when user confirmed intake.
- `status` (Enum: `MedicineStatus`): `pending`, `taken`, `skipped`, `missed`, `snoozed`.
- `skipReason` (String?): Predefined choice (`"Nausea"`, `"Doctor Advised"`, `"Felt Fine"`, `"Ran Out"`, `"Other"`).
- `notes` (String?): Optional user observation.

6.4 Entity Specification: `EmergencyContact`

- `id` (Id): Primary key.
- `fullName` (String): Name of doctor, clinic, or caregiver.
- `relationship` (String): `"Primary Care Doctor"`, `"Cardiologist"`, `"Spouse"`, `"Son/Daughter"`.
- `phoneNumber` (String): Clean phone digits for direct telephony intents.
- `isPrimary` (bool): Highlighted emergency speed-dial recipient.

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7. Core Service Layer Contracts

All services are registered as singletons in `lib/app/locator.dart`.

```dart
final locator = GetIt.instance;

void setupLocator() {
  locator.registerLazySingleton<DatabaseService>(() => DatabaseService());
  locator.registerLazySingleton<NotificationService>(() => NotificationService());
  locator.registerLazySingleton<AlarmService>(() => AlarmService());
  locator.registerLazySingleton<LocalStorageService>(() => LocalStorageService());
  locator.registerLazySingleton<PermissionService>(() => PermissionService());
}
```

7.1 `DatabaseService` (Local Isar Data Access Object)

- `Future<void> init()`: Opens Isar instance, handles schema migrations.
- `Future<List<Medicine>> getAllMedicines()`: Streams or fetches all active medications.
- `Future<int> saveMedicine(Medicine medicine, List<ReminderTime> reminders)`: Saves medication and schedules reminders atomically.
- `Future<void> deleteMedicine(int id)`: Cascades deletion to reminders and clears scheduled alarms.
- `Future<List<DoseLog>> getDoseLogsForDate(DateTime date)`: Returns all intake entries for a specific day.
- `Future<void> logDoseIntake(int doseLogId, MedicineStatus status, {String? reason})`: Updates intake log and decrements medication inventory if taken.
- `Future<AdherenceStats> getAdherenceStats(DateTime startDate, DateTime endDate)`: Computes adherence rate, taken count, missed count, and streak.
- `Future<String> exportDatabaseToJson()`: Serializes all collections to a JSON string.
- `Future<bool> importDatabaseFromJson(String jsonContent)`: Validates and restores database state.

7.2 `AlarmService` & `NotificationService`

- `Future<void> scheduleMedicationAlarm(Medicine medicine, ReminderTime reminder)`: Registers high-priority exact alarm via `flutter_local_notifications` and Android `AlarmManager`.
- `Future<void> cancelMedicationAlarm(int reminderId)`: Removes scheduled system trigger.
- `Future<void> triggerFullScreenAlarm(int medicineId, int reminderTimeId)`: Displays full-screen lockscreen-overriding alert with audio loop and vibration.
- `Future<void> snoozeAlarm(int doseLogId, int durationMinutes)`: Reschedules trigger after specified minutes.

7.3 `LocalStorageService` (`SharedPreferences` Wrapper)

- `bool get isFirstTimeUser` / `setFirstTimeUser(bool val)`
- `bool get isBiometricLockEnabled` / `setBiometricLockEnabled(bool val)`
- `String? get pinCodeHash` / `setPinCodeHash(String hash)`
- `ThemeMode get themeMode` / `setThemeMode(ThemeMode mode)`
- `int get defaultSnoozeMinutes` / `setDefaultSnoozeMinutes(int mins)`

7.4 `PermissionService`

- `Future<bool> requestNotificationPermission()`: Requests system notification rights.

- `Future<bool> requestExactAlarmPermission()`: Requests Android 12+ `SCHEDULE_EXACT_ALARM`.
- `Future<bool> requestCameraAndGalleryPermission()`: Requests camera and photo storage access for pill photos.
- `Future<bool> authenticateWithBiometrics()`: Evaluates Face ID / Fingerprint via `local_auth`.

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8. Detailed Screen Specifications & User Flows

```
+------------------------------------------------------------------------+
|                            APP NAVIGATION FLOW                         |
|                                                                        |
|  [ Splash Screen ] ---> [ Permission / Onboarding ]                    |
|           |                                                            |
|           v                                                            |
|  [ PIN / Biometric App Lock ] (If enabled)                             |
|           |                                                            |
|           v                                                            |
|  +------------------------------------------------------------------+  |
|  | Main Shell (Bottom Navigation Bar)                               |  |
|  |  (1) Dashboard   (2) History/Calendar  (3) Reports  (4) Settings |  |
|  +------------------------------------------------------------------+  |
|        |                  |                                            |
|        +---> [ + FAB ] -> [ Add / Edit Medication Wizard ]             |
|        |                                                               |
|        +---> [ Alarm Event Triggered ] -> [ Full-Screen Alarm Modal ]  |
|        |                                                               |
|        +---> [ Emergency Icon ] -> [ Emergency Speed-Dial Sheet ]     |
+------------------------------------------------------------------------+
```

8.1 Splash & App Startup Sequence

1. `WidgetsFlutterBinding.ensureInitialized()` executes.
2. `setupLocator()` registers singletons.
3. `LocalStorageService.init()` loads stored flags.
4. `DatabaseService.init()` opens Isar local storage.
5. `NotificationService.init()` configures notification channels and exact alarm permissions.
6. **Routing Decision:**
   - If `isFirstTimeUser == true` → Navigate to `OnboardingScreen`.
   - If `pinCodeHash != null` → Navigate to `AppLockScreen`.
   - Otherwise → Navigate to `DashboardScreen`.

8.2 Onboarding Flow (`OnboardingScreen`)

- **Slide 1:** “Never Miss a Dose” — Visual alarm reminders with exact timing.
- **Slide 2:** “Visual Pill Verification” — Attach pill photos and food instructions.
- **Slide 3:** “100% Offline & Private” — No accounts, no servers, full health privacy.
- **Action:** Permission grant step (Notifications + Alarms) → “Get Started” CTA.

8.3 Dashboard (`DashboardScreen`)

- **Header:** Personalized greeting, current date, and Emergency Speed-Dial icon.
- **Adherence Summary Card:** Circular SVG/Canvas progress ring showing today's score: `X of Y Doses Taken` (with motivational text, e.g., "Great job! You're on track today").
- **Week Strip / Horizontal Date Selector:** Tappable 7-day strip allowing navigation to past days or tomorrow.
- **Segmented Timeline View:** Doses grouped chronologically:
  - **Morning** (05:00 – 11:59)
  - **Afternoon** (12:00 – 16:59)
  - **Evening** (17:00 – 21:59)
  - **Night** (22:00 – 04:59)
- **Timeline Medication Card:**
  - Left: Circular pill photo or color-coded icon avatar.
  - Middle: Medicine Name (`HeadlineSmall`), Dosage (`500 mg - 1 Tablet`), Meal Chip (`After Meal`).
  - Right: Scheduled Time Badge + Status Indicator (Green Check for Taken, Red Exclamation for Missed).
  - Bottom Quick Actions: Full-width **[Mark as Taken]**, **[Snooze 10m]**, **[Skip]**.
- **Floating Action Button (FAB):** Prominent Center-docked primary teal + button to add new medication.
- **Empty State:** Illustrated soothing graphic with "No medications scheduled for today" and "Add Your First Medication" button.

8.4 Add / Edit Medication Workflow (`AddMedicineScreen`)

A structured 4-step wizard designed to eliminate cognitive fatigue: - **Step 1: Medicine Identification** - Text input for Medicine Name (with instant duplicate name warning prompt). - Photo Attachment: Camera capture or Gallery picker; stored locally via `path_provider`. - Form factor icon selector (Tablet, Capsule, Liquid, Injection, Inhaler, Drops). - Color palette accent picker for UI identification. - **Step 2: Dosage & Meal Relation** - Numerical Dosage input (e.g., `250`, `500`) + Unit selector dropdown (`mg`, `ml`, `pills`, `units`). - Meal relation chips: `[Before Meal]`, `[After Meal]`, `[With Food]`, `[Empty Stomach]`, `[No Relation]`. - **Step 3: Schedule & Alarm Times** - Frequency selector: `Daily`, `Specific Days of Week` (M T W T F S S toggles), `Hourly Interval` (Every 4/6/8/12 hrs), or `As Needed (PRN)`. - Alarm Times: Add multiple time triggers per day (e.g., `08:00 AM`, `02:00 PM`, `08:00 PM`). - Course Duration: Toggle `Ongoing Maintenance` or set `Start Date` → `End Date`. - **Step 4: Inventory & Extra Notes (Optional)** - Current Pill Stock count (e.g., 30) + Low Stock Alert Threshold (e.g., 5 pills left). - Doctor Name & Prescription Notes (e.g., "Prescribed by Dr. Smith for ear infection"). - **Validation Rules:** - Name and dosage cannot be empty or zero. - At least one alarm time must be configured for non-PRN schedules. - Saving persists to Isar and registers exact alarms immediately.

8.5 Full-Screen Active Alarm Screen (`ActiveAlarmScreen`)

Launched upon exact alarm trigger with continuous high-volume audio and vibration (overriding device lockscreen): - **Hero Image:** High-resolution display of the user's custom pill photo. - **Critical Labels:** High-contrast Medicine Name (`DisplayLarge`), Exact Dosage, and prominent Meal Guidance Badge (e.g., *"Take with a full meal"* (warning badge)). - **Three High-Contrast Action Triggers: - [Mark as Taken]** (green) (Huge, prominent touch button) → Stops audio, logs `Taken`, decrements stock, closes screen. - **[Snooze (10 Mins)]** (yellow) → Schedules snooze timer, dismisses screen. - **[Skip Dose]** (red) → Opens modal sheet with reasons (`Felt fine`, `Side effects`, `Doctor advice`, `Ran out`), logs `Skipped`. - **Auto-Timeout Safety Rule:** If unattended for 30 minutes, system stops audio and automatically marks dose as `Missed`.

8.6 Adherence History & Calendar View (`HistoryScreen`)

- **Interactive Calendar Heatmap:**
  - Color-coded day cells: Green (100% adherence), Yellow ($50\% - 99\%$), Red ($< 50\%$ or missed doses), Grey (No scheduled doses).
  - Tapping a calendar date filters the dose log list below.
- **Dose History List:**
  - Filter chips: `[All]`, `[Taken]`, `[Skipped]`, `[Missed]`.
  - Log item shows time taken vs. time scheduled.
  - **Retroactive Correction:** Allows user to tap any past dose and manually update status (e.g., "I took this 30 mins ago").

8.7 Reports, Analytics & PDF Generation (`ReportsScreen`)

- **Metric Cards:**
  - Overall Monthly Adherence Percentage (`94%`).
  - Total Doses Taken vs. Missed vs. Skipped.
  - Current Consecutive Intake Streak (e.g., `14 Days`).
- **Visual Adherence Charts (`fl_chart`):**
  - Donut Chart: Breakdown of intake statuses for the selected month.
  - Weekly Bar Chart: Day-by-day compliance comparison.
- **Printable Medical PDF Export:**
  - Generates a clinical-grade PDF summary using `pdf` & `printing` packages.
  - PDF includes: Patient Name, Reporting Period, Total Adherence %, Medication List with Dosages, Detailed Log Table, and Doctor Signature line.
  - Direct actions: **[Preview PDF]**, **[Print]**, **[Share / Save to Files]**.

8.8 Emergency Contacts (`EmergencyScreen`)

- Displays list of doctors, specialists, and family caregivers.
- 1-Tap Speed-Dial Button: Fires direct OS phone dialer (`tel:1234567890`) via `url_launcher`.
- 1-Tap SMS Alert: Fires SMS intent with pre-filled message ("Hello, I need assistance with my medication").

8.9 Settings, Security & Data Management (`SettingsScreen`)

- **App Security:**
  - Toggle 4-digit PIN lock.
  - Toggle Biometric authentication (Face ID / Fingerprint).
- **Alarm Preferences:**
  - Custom Alarm Ringtone selector.
  - Vibration pattern toggle.
  - Default Snooze duration slider (`5`, `10`, `15`, `30` mins).
  - Missed dose grace period slider (`15`, `30`, `60` mins).
- **Theme Selection:**
  - `Light Mode` (Clinical Humanist default), `Dark Mode`, `System Default`.
- **Local Data Backup & Recovery:**
  - **[Export Database Backup (JSON)]**: Writes complete database dump to device storage.
  - **[Restore Database from JSON]**: Validates JSON schema, restores collections, and reschedules all active alarms.
  - **[Clear All Data]**: Destructive reset with double confirmation modal.

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9. Non-Functional Requirements & Engineering Standards

9.1 Performance & Reliability

- **Cold App Launch:** < 1.5 seconds to interactive dashboard.
- **Database Query Latency:** < 15 ms for timeline queries on datasets with $10,000+$ dose logs.
- **Frame Rate:** Consistent 60 FPS scrolling on all lists and animations.
- **Memory Footprint:** < 120 MB RAM during active execution.
- **Alarm Reliability:** 99.99% execution rate using Android `SCHEDULE_EXACT_ALARM` with Foreground Service and iOS native local notifications.

9.2 Data Privacy & Compliance

- **Zero Telemetry / Tracking:** No third-party analytical trackers, crash SDKs transmitting PII, or remote logging.
- **Local File Sandboxing:** Pill images stored in application private directory (`getApplicationDocumentsD` preventing indexing by generic photo galleries unless exported.

9.3 Accessibility (a11y) & Usability Standards

- **WCAG 2.1 AA Compliance:** Minimum color contrast ratio of $4.5:1$ for normal text and $3.0:1$ for large text and UI components.
- **Touch Target Size:** Minimum hit target of 48×48 dp across all interactive widgets.
- **Screen Reader Support:** Semantic labels applied to all icon buttons and photo cards (`Semantics(label: "Take Amoxicillin 500mg")`).

---

10. Verification & Quality Assurance Plan

```
+--------------------------------------------------------------------+
|                          AUTOMATED TESTING SUITE                    |
|                                                                     |
|  [ Unit Tests ]          -> Adherence formulas, Isar DAOs, ViewModels   |
|  [ Widget Tests ]        -> Timeline cards, Progress ring, Alarm dialog |
|  [ Integration Tests ]   -> Full Add Medicine -> Alarm Trigger flow    |
+--------------------------------------------------------------------+
```

10.1 Automated Test Suites

1. **Adherence Formula Unit Tests:**
   - Verify calculation when skipped doses > 0.
   - Verify streak reset logic upon missed dose.
2. **Database DAO Unit Tests:**
   - Test CRUD operations for `Medicine`, `ReminderTime`, and `DoseLog`.
   - Test cascading deletion of reminders when a medicine is deleted.
   - Test JSON export and import data integrity.
3. **ViewModel State Unit Tests:**
   - Verify `DashboardProvider.loadTodayTimeline()` transitions from `ViewState.busy` to `ViewState.idle`.
   - Verify error state handling when storage is full.

4. **Widget Tests:**
    - Validate responsive sizing on small screens (320dp) and large tablets (600dp+).
    - Test that "Mark as Taken" updates card state without full screen redraw glitch.

10.2 Manual QA Checklist

- ☐ Test alarm firing when device is locked and screen is off.
- ☐ Test alarm firing when device is in "Do Not Disturb" (verifying high-priority alarm channel override).
- ☐ Test app kill/reboot scenario: Ensure alarms are rescheduled on `BOOT_COMPLETED`.
- ☐ Test pill image capture from camera and verify thumbnail displays correctly on active alarm screen.
- ☐ Test PDF export generation and verify layout on physical printout.
- ☐ Test PIN lock protection upon backgrounding and resuming app.

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11. CI/CD Pipeline & Git Branching Strategy

```
dev (Active Development & Feature Branches)
 │
 ├──> PR & Code Review
 │
 ▼
test (Automated CI Builds, Unit & Integration Tests)
 │
 ├──> Build APK / AppBundle (GitHub Actions: build_apk_appbundle.yml)
 │
 ▼
prod (Tagged Production Releases: v1.0.0+1)
```

- **Branch Conventions:**
    - `dev`: Active development; all feature pull requests merge here.
    - `test`: Automated CI branch; triggers Android build & test runner.
    - `prod`: Production-ready release branch; uploads to Google Play Internal / TestFlight.
- **Versioning Strategy:**
    - Controlled via `pubspec.yaml`: `version: MAJOR.MINOR.PATCH+BUILD_NUMBER` (e.g., `1.0.0+1`).

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12. Implementation Roadmap & Milestones

Milestone	Target Deliverables	Expected Timeline
M1: Foundation & Local DB	Set up Isar collections, DAOs, Service Locator, `AppTheme`, and `ScreenUtil`.	Sprint 1
M2: Medicine CRUD & Storage	Build Add/Edit Medication Wizard, Pill Photo Picker, and Local Storage.	Sprint 2
M3: Alarm & Notification Engine	Implement exact alarms, full-screen lockscreen alert, and Snooze/Skip handlers.	Sprint 3

Milestone	Target Deliverables	Expected Timeline
M4: Dashboard & History	Implement Timeline UI, Daily Progress Ring, and Calendar Heatmap.	Sprint 4
M5: Analytics & PDF Reports	Build `fl_chart` analytics, medical PDF generator, and JSON backup/restore.	Sprint 5
M6: Security, Polish & QA	Implement PIN/Biometrics, Emergency Speed-Dial, UI accessibility, and full QA.	Sprint 6

MediAlert Product Requirements Document — Version 2.0 (Offline-First Edition)